

Objects [SimTalk] - SankeyDiagram

Sets the objects, i.e., the *Parts* or *Workers* whose *Sankey* flows which the *SankeyDiagram* designated by `<Path>` visualizes in the *Frame*.

Type

Attribute

Syntax

```
<Path>.Objects:array
```

Assignment Value

You can assign a value of data type *array*.

Example

```
MySankeyDiagram.Objects := [.Resources.Worker]           // all Workers of
this class
MySankeyDiagram.Objects := [.Resources.WorkerPool]       // all WorkerPool
classes
MySankeyDiagram.Objects := [.Models.Model.WorkerPool]    // all Workers of
the WorkerPool in the Frame named Frame
MySankeyDiagram.Objects := [.Resources.Worker:1]         // the Worker with
the number 1
```

See also

[Objects](#), tab

CostAnalyzer

CostAnalyzer

Use the object *CostAnalyzer* for analyzing costs that the individual *machines* and the *Worker* cause. The *CostAnalyzer* computes the costs that accrue for the material flow objects while processing the parts in the facility.

Image

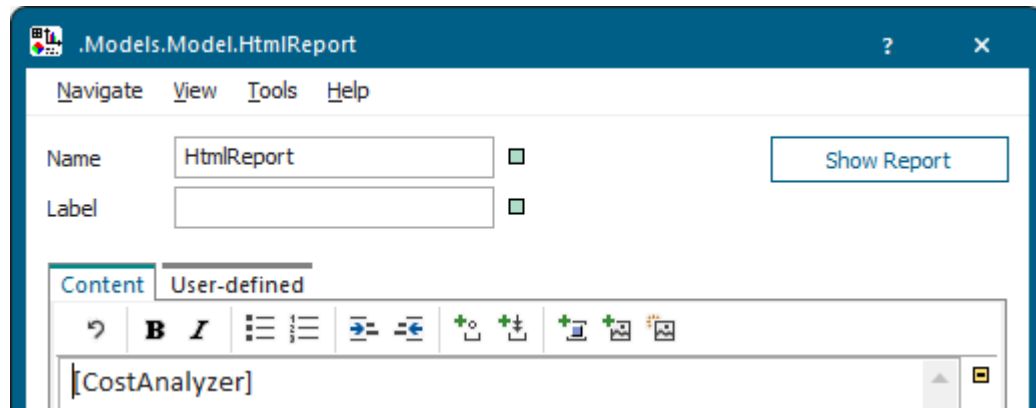


Description

The *CostAnalyzer* also computes the summed up investment costs of the objects. You cannot configure the *CostAnalyzer* as such.

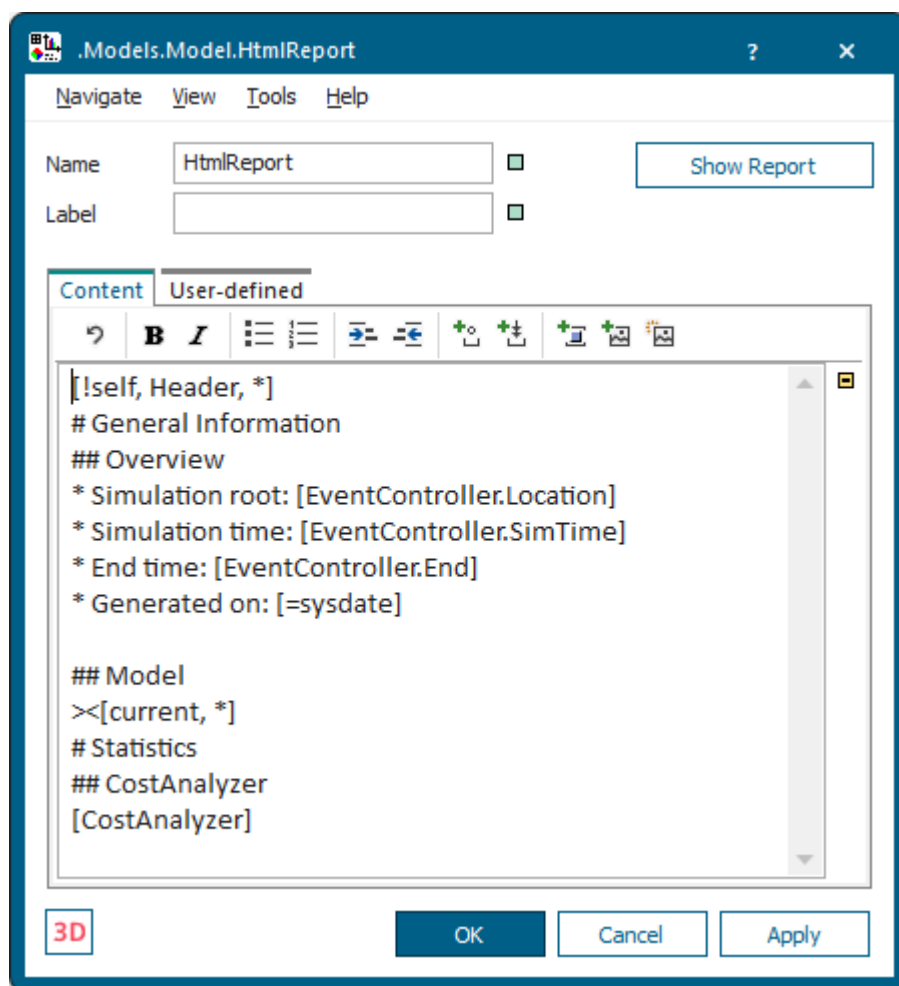
To show the results of the cost analysis, **drag** the *CostAnalyzer* from the *Frame* onto the icon of the *HtmlReport* and **drop** it there.

- If the **Content** of the *HtmlReport* is inherited, the data of the *CostAnalyzer* replaces the pre-defined content of the *HtmlReport*. *Plant Simulation* opens the display window of the *HtmlReport* automatically.

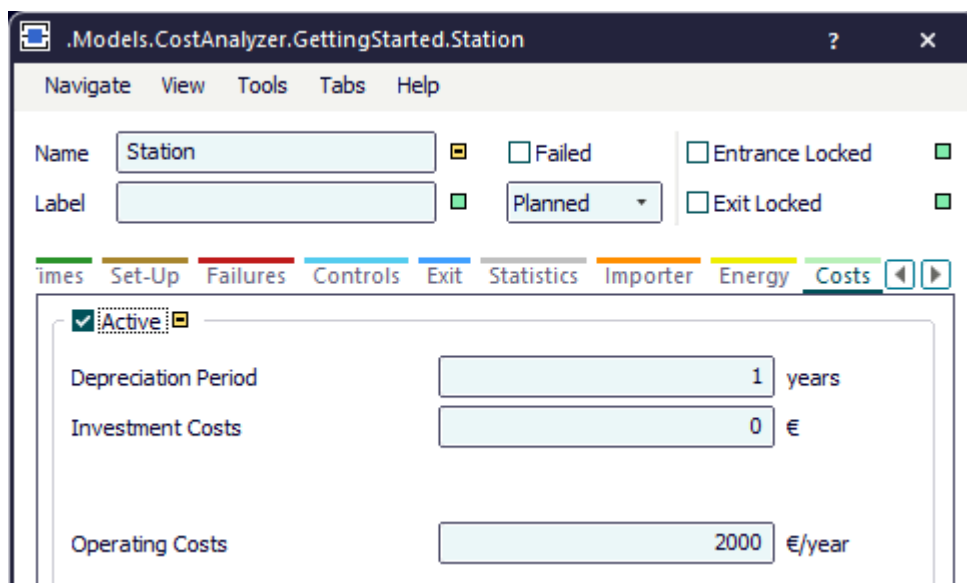


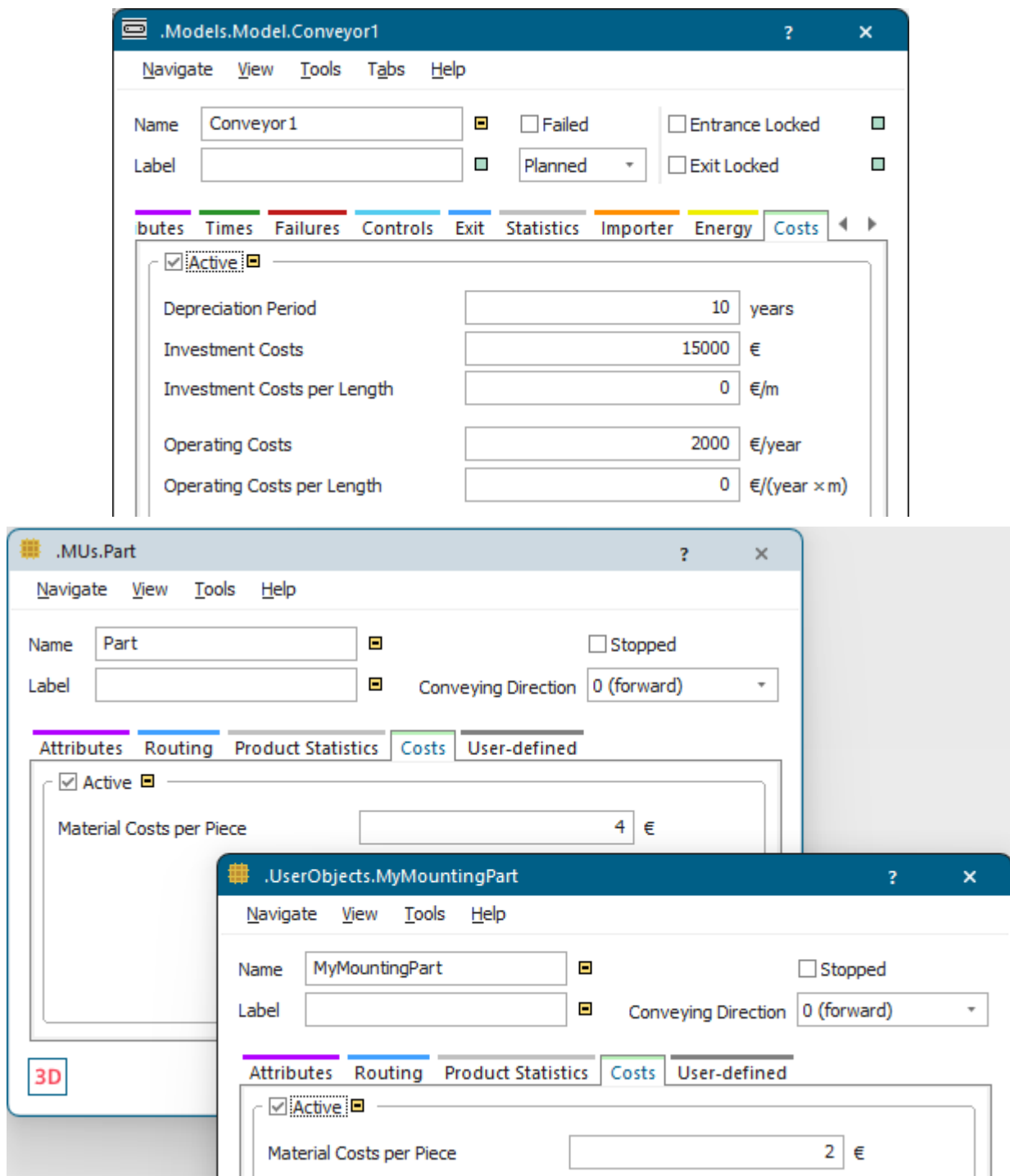
- If the **Content** of the *HtmlReport* is not inherited, *Plant Simulation* appends the data of the *CostAnalyzer* to the pre-defined content of the *HtmlReport*. *Plant Simulation* opens the display window of the *HtmlReport* automatically.

To show the results of the analysis, you can also drag the object *CostAnalyzer* into the open window of the *HtmlReport* and drop it there, for example below the heading **Statistics**. Then click **Show Report**.



Type in the respective costs on the tab **Costs** of the *active material flow objects* and the tab **Costs** of the *Part* and of the *Container*.



**Note**

At the moment you can only set if the *CostAnalyzer* collects data or not. We do not provide any other configuration settings.

To show a **tooltip** with information about the *CostAnalyzer*, hover with the mouse over it.

To change the length of the graphic and the anchor points of the *CostAnalyzer*, click [Show Manipulators](#)



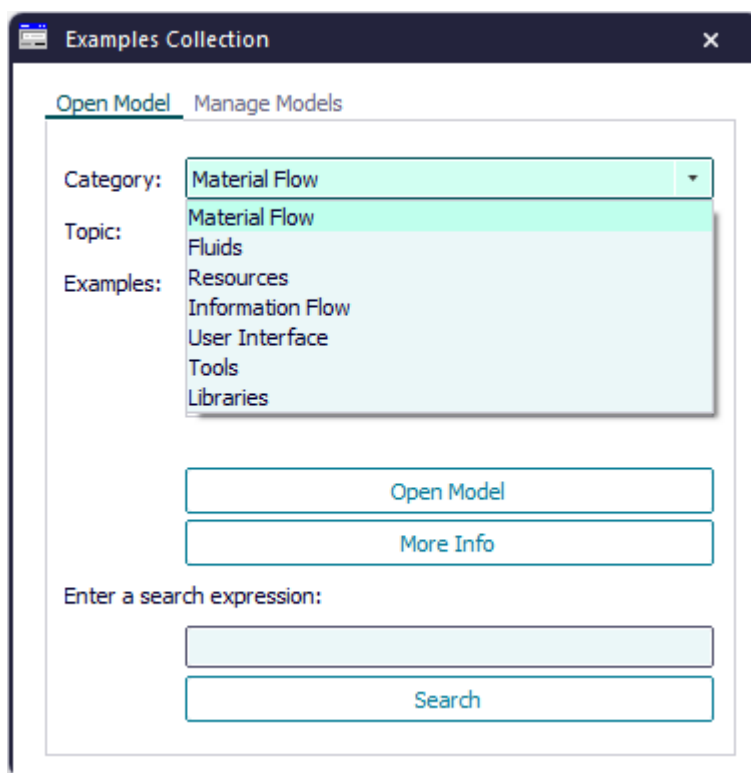
on the **Edit** ribbon tab or press **M** on the keyboard.

Add the Object to the Simulation Model



To add the object *CostAnalyzer* to your simulation model, click [Manage Class Library](#) > **Basic Objects** > **UserInterface** > **CostAnalyzer** on the **Home** ribbon tab.

Compare the sample models: Click the **Window** ribbon tab, click **Start Page** > **Getting Started** > **Example Models** > **Small Examples**. Then, select the respective **Category**, the **Topic**, and the **Example** in the dialog **Examples Collection**, and click **Open Model**.



See also

[Simulate the Accrued Costs of the Machines](#)

[How the CostAnalyzer Assigns Costs to Part Types](#)

[Costs Shown in the Costs Report](#)

How the CostAnalyzer Assigns Costs to Part Types

The material flow objects generate costs during the simulation which result from the total investment costs and the total operating costs per time. These costs are distributed as accrued costs to the parts which the material flow objects process.

Remarks

If no parts are located on the material flow objects, the costs are allocated to the respective material flow object as general costs.

If the *CostAnalyzer* computes the result, these general costs are distributed across the part types which the material flow object processed during the simulation. Parts with the same name are allocated to the same part type (**MU Type**). *Plant Simulation* uses the name that the part has when it arrives at the *Drain*.

Note

For the **Work in process** the final production costs are not known. For this reason they only contribute with 50 % to the production costs. The assumption is that the work in process parts have covered about half of their production process on average.

Costs for parts which are currently passing through the production process are not precise. Piece costs are only then meaningful if the **Throughput** is considerably higher than the **Work in process**.

Not all material flow objects proceed the same during this process.

- The **AssemblyStation** assigns the costs to the main part. Costs, which accrued for the mounting parts until then, are transferred to the main part.
- The **Conveyor** assigns the costs proportionately according to the length of the part.
- The **DismantleStation** assigns the costs to the main part.
- The **ParallelStation** distributes the costs evenly across its processing places.
- The **Container** distributes the accrued costs evenly across the storage places if **Distribute costs** is activated.

In addition, material costs accrue during the simulation when a new part enters the production process.

See also

[Depreciation Period \[text box\]](#)

[Investment Costs \[text box\]](#)

[Investment Costs per Length \[text box\]](#)

Operating Costs [text box]

Operating Costs per Length [text box]

Material Costs per Piece [text box]

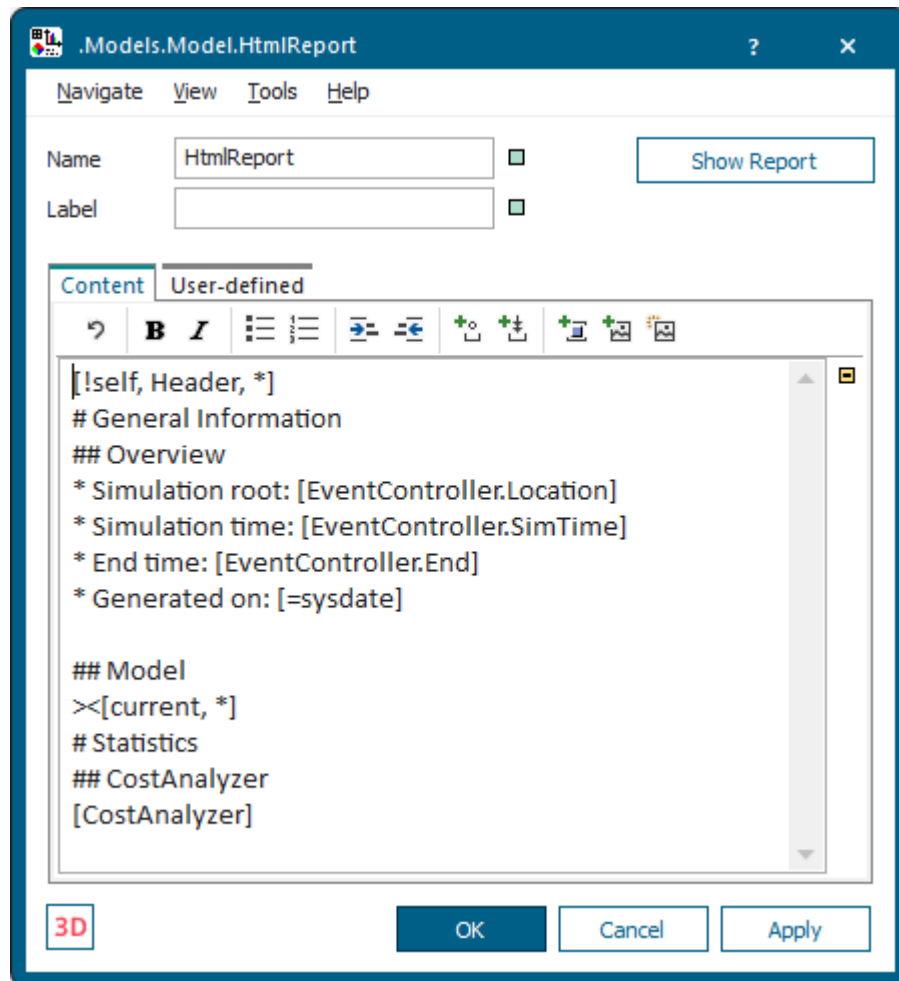
Distribute Costs [check box]

Costs Shown in the Costs Report

The *HtmlReport* shows the tables **Investment Costs** and **Piece Costs** of the cost analysis by default.

Remarks

You cannot configure the object *CostAnalyzer* as such. To show the results of the analysis, drag the object *CostAnalyzer* into the open window of the *HtmlReport* and drop it there, for example below the heading **Statistics**. Then click **Show Report**.



The result looks like this:

The screenshot shows a web browser window titled ".Models.Model.HtmlReport". On the left is a sidebar with a tree view containing "General Information", "Statistics", and "CostAnalyzer". The main content area is titled "Statistics" and "CostAnalyzer". It displays a table with investment and operating costs for various model objects.

Object	Investment costs [€]	Depreciation period [years]	Operating costs [€/year]
Model	25000.00		5750.00
└ SubFrame	8500.00		600.00
└└ Conveyor	2000.00	8.00	200.00
└└ Station	4500.00	10.00	200.00
└└ Conveyor1	2000.00	10.00	200.00
└ AssemblyStation	0.00	1.00	1000.00
└└ Conveyor	1500.00	8.00	150.00
└└ Station	0.00	1.00	2000.00
└└ Conveyor1	15000.00	10.00	2000.00

Below the table is the text "Investment costs".

MU Type	Piece costs [€]	Material costs [€]	Accrued costs [€]	General costs [€]	Throughput	Work in process
Part	8.32	8.00	0.17	0.15	10650	6

Below this table is the text "Piece costs".

At the bottom of the sidebar, it says "Plant Simulation 2504 Siemens Digital Industries Software".

Compare [View the Costs Report](#) in the *Step-by-Step help*.

Investment Costs

The table **Investment Costs** shows the **investment costs**, the **depreciation period**, and the **operating costs** of the material flow objects inserted into the model hierarchically structured.

Note

The *HtmlReport* shows the investment costs statically, i.e., the values do not change during the simulation run.

For *Frames* in the hierarchy the table shows the summed up investment costs. These costs are highlighted in light gray. The first row shows the summed up investment costs of the entire model.

For *sub-Frames* in the hierarchy the table shows the summed up investment costs as well. These costs are highlighted in light gray. The first row shows the summed up investment costs of this *sub-Frame*.

SimTalk

[putInvestmentCostsIntoTable \[SimTalk\]](#)

[getInvestmentCostsTable \[SimTalk\]](#)

Piece Costs

The table **Piece Costs** shows the costs which accrued for the part types in the individual rows during the simulation.

The **MU Type** (part type) designates the name of the parts when they arrive in the *Drain*. When parts change their name during the simulation run, for example during the different production phases, the costs are assigned to the final product.

The **Material costs** designate the costs which accrue during the simulation when a new part enters the production.

The **Accrued costs** designate the costs which are assigned to the part while it is processed on the material flow object.

The **General costs** designate the costs which accrue when a material object is empty during the simulation. These costs are distributed across the part types which were processed by the material flow object.

The **Piece costs** is the sum of the **material costs**, the **accrued costs**, and the **general costs**.

The **Throughput** designates the number of parts which were processed and moved through the production line.

The **Work in process** designates the number of parts which are currently passing through production process. These parts only contribute by half for computing the piece costs.

All costs are averaged per piece.

Note

For the **Work in process** the final production costs are not known. For this reason they only contribute with 50 % to the production costs. The assumption is that the work in process parts have covered about half of their production process on average.

Costs for parts which are currently passing through the production process are not precise. Piece costs are only then meaningful if the **Throughput** is considerably higher than the **Work in process**.

SimTalk

[getPieceCostsTable \[SimTalk\]](#)

[putPieceCostsIntoTable \[SimTalk\]](#)

See also

[Depreciation Period \[text box\]](#)

[Investment Costs \[text box\]](#)

[Investment Costs per Length \[text box\]](#)

[Operating Costs \[text box\]](#)

[Operating Costs per Length \[text box\]](#)

[Material Costs per Piece \[text box\]](#)

[Distribute Costs \[check box\]](#)

Dialog Box of the Cost Analyzer

Dialog Box of the CostAnalyzer

Double-click the icon of the *CostAnalyzer* to open its dialog box.

Edit Simulation Properties

In the dialog box you can change the **simulation properties** of the object. The shared properties are described under [Dialog Items of the Objects](#).

Edit Animation Properties

To edit the **3D properties** of the object in the dialog box [Edit 3D Properties](#):

- Click the button  **Edit 3D Properties** in the lower left corner of the simulation properties dialog box.
- Select the object in the model and press the **spacebar**.

To manipulate the graphic of the object, click [Show Manipulators](#) on the **Edit** ribbon tab or press **M** on the keyboard.

See also

[Collect Data \[check box\] - CostAnalyzer](#)

[Investment Costs \[button\]](#)

[Piece Costs \[button\]](#)

Collect Data [check box] - CostAnalyzer

To make the *CostAnalyzer* collect data during the simulation run, select this check box. To deactivate data collection, clear the check box.

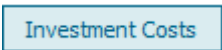
☒ Collect Data ☐

SimTalk

CollectData [SimTalk] - CostAnalyzer

Investment Costs [button]

To open a list, which contains the **investment costs**, which the *CostAnalyzer* computed, click this button.



.Models.Model.CostAnalyzer.getInvestmentCostsTable				
	Object	Investment costs	Depreciation period	Operating costs
1	*.Models.Model	25000.00		5750.00
2	*.Models.Model.SubFrame	8500.00		600.00
3	*.Models.Model.SubFrame.Conveyor	2000.00	8.00	200.00
4	*.Models.Model.SubFrame.Station	4500.00	10.00	200.00
5	*.Models.Model.SubFrame.Conveyor1	2000.00	10.00	200.00
6	*.Models.Model.AssemblyStation	0.00	1.00	1000.00
7	*.Models.Model.Conveyor	1500.00	8.00	150.00
8	*.Models.Model.Station	0.00	1.00	2000.00
9	*.Models.Model.Conveyor1	15000.00	10.00	2000.00

SimTalk

[getInvestmentCostsTable \[SimTalk\]](#)

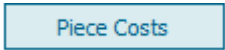
See also

[View the Costs Report](#)

Piece Costs [button]

Piece Costs [button]

To open a list, which contains the **piece costs**, which the *CostAnalyzer* computed, click this button.



.Models.Model.CostAnalyzer.getPieceCostsTable							
	MU Type	Piece costs	Material costs	Accrued costs	General costs	Throughput	Work in process
1	Part	8.32	8.00	0.25	0.06	53114	24

SimTalk

[getPieceCostsTable \[SimTalk\]](#)

See also

[View the Costs Report](#)

Tab User-defined

Define your own attributes as described under the [Tab User-defined](#).

Navigate Menu

The commands are described under the [Navigate Menu](#).

View Menu

The commands are described under the [View Menu](#).

SimTalk

[updateDialog \[SimTalk\]](#)

Tools Menu

The **Tools Menu** provides commands to access its functions.

[Edit Controls](#)

[Edit Observers](#)

See also

[Tools Menu \[general description\]](#)

Help Menu

The commands are described under the [Help Menu](#).

Methods of the CostAnalyzer

_Methods of the CostAnalyzer

The *CostAnalyzer* provides;

- The *methods* listed in the table of contents to the left.
- The [Methods of All Objects](#).

To view all of the *methods*, *read-only attributes*, and *attributes* of the object, open the window [Show Attributes and Methods](#). The figure below illustrates the information using the example of the object *Station*.

Name	Value	Inherited	Watchable	Signature
contentsList	[]			([Contents:table]) -> any
createAttr				(NameOfUserDefinedAttribute:string, DataType:string) -> boolean
createIcon				(IconName:string, Width:integer, Height:integer) -> integer
deleteAttr				(NameOfUserDefinedAttribute:string) -> boolean

Name of the method Identifier and data type of the parameter you enter Data type of the return value

Name	Value	Inherited	Watchable	Signature
moveToFolder				(Folder:object) -> object
MU	VOID			([Index:integer:=1]) -> object
MUPart	VOID			([Index:integer:=1]) -> object

Name of the method Identifier and data type of the parameter you enter Default value Data type of the return value

- Select **Show Attributes and Methods** on the context menu of the **Class Library** to show the *methods*, *read-only attributes*, and *attributes* of the selected [Class \[general description\]](#).
- Press the **F8** key or click **Show Attributes and Methods** on the **Home** ribbon tab of the *Frame* into which you inserted an instance to show the *methods*, *read-only attributes*, and *attributes* of the selected [Instance \[general description\]](#).